



Kulpio

Waste less food. Save money. In any language.

A privacy-first Progressive Web App I designed and built to help households waste less food — offline, on any phone, in 33 languages. Built for the UN Global Goals.

FIRSTSTEP PROJECT COMPETITION 2026

CREATIVE ENGINEERS

Daniil Bejenari

T.L. Orizont · Ceadâr-Lunga

WHY I BUILT THIS

It started in my own fridge



I kept watching good food go bad — bought with the best intentions, pushed to the back of the shelf, forgotten, then thrown out.

That bothered me twice over: it's money in the bin, and wasted food is a real climate problem. **So I built the tool I wanted** — one that just tells me what to use next, before it's too late.

The idea in one line

“Don't let me forget the food I already own.”

THE PROBLEM

A third of our food is wasted — mostly at home

~1/3

of all food produced is lost or wasted (FAO)

#1

households are among the largest single contributors

Forget

the main cause at home:
bought, forgotten, expired



It's an information & attention problem — not a preservation problem.
That's exactly what an app can fix.

Built for the Sustainable Development Goals



PRIMARY

Responsible Consumption & Production

Kulpio's core mission — cut the food households throw away.



Zero Hunger

Use more of the food we buy; value it instead of binning it.



Climate Action

Less wasted food means fewer needless food-related emissions.



UN Target 12.3: halve per-capita global food waste by 2030. Kulpio goes straight at household waste — the largest, most-forgettable slice.

THE IDEA

Track it, know what's going off, cook it first

01



Add in seconds

Scan a barcode, read a label or receipt with AI, or paste a list — no typing marathon.

02



Know what expires

Printed dates beat estimates. Live freshness colours and a 'week-ahead' strip.

03

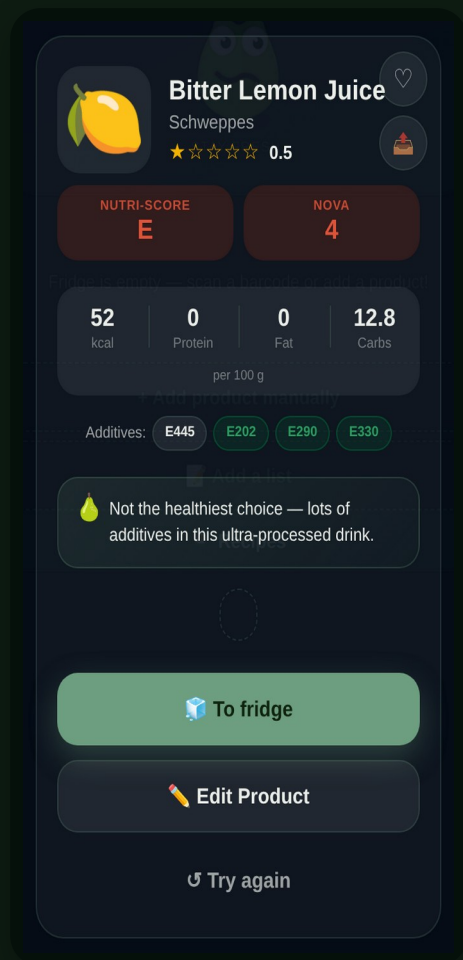


Use it in time

Cook what's going off first; mark Used / Wasted / Froze and watch the money you save.

SEE IT

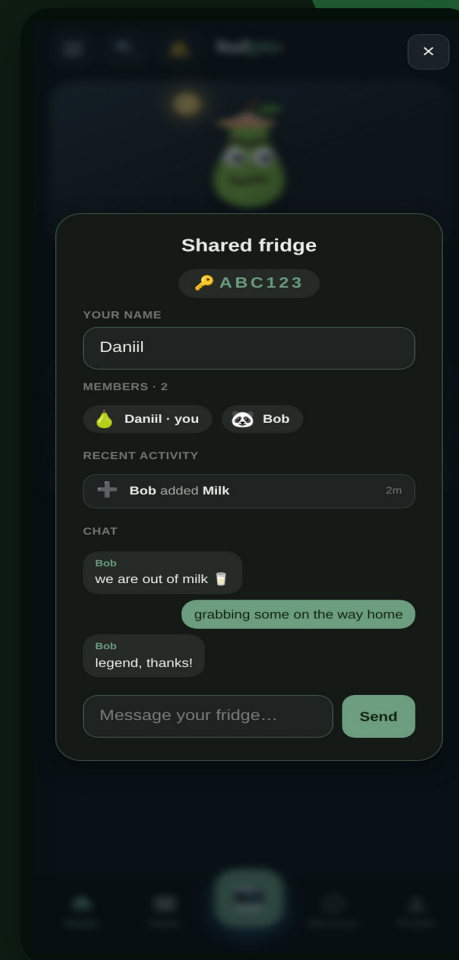
One app, offline, on any phone



Scan → product card
additives, Nutri-Score, verdict



Your impact
waste cut & money saved



Shared fridge
members, activity & chat

FEATURES

What Kulpio does



Effortless input

Barcode, AI label & receipt scan, or paste a list



Storage sections

Fridge, freezer & pantry with live freshness



Recipes

Cook from your ingredients, with nutrition estimates



Shared fridge

One code links a household + activity feed & chat



Motivation

Achievements, XP levels, monthly recap & impact



33 languages

Full RTL support; installs to the home screen

METHODOLOGY

How Kulpio decides when food expires

Printed dates are ground truth — Kulpio only estimates when there's no date to read.

1

Printed date wins

Read from the pack (OCR + model) or typed by you. Always takes priority.

2

Offline shelf-life table

Built-in per-food estimate, adjusted for storage (freezer + ~90 d; opened shortens).

3

AI refine

When online, the model refines the estimate from the product name.

AI degrades gracefully

Best cloud AI model (top quality)



Cloudflare Workers AI — Llama (free)



Clean error — never blocks

Built as one offline file + one worker



Client

- **Single static HTML** — ~1.3 MB — markup, CSS, JS, 33 langs
- **No build step** — no framework, no bundler
- **Service worker** — precache → fully offline
- **Web Crypto** — hashing & secure tokens



Backend

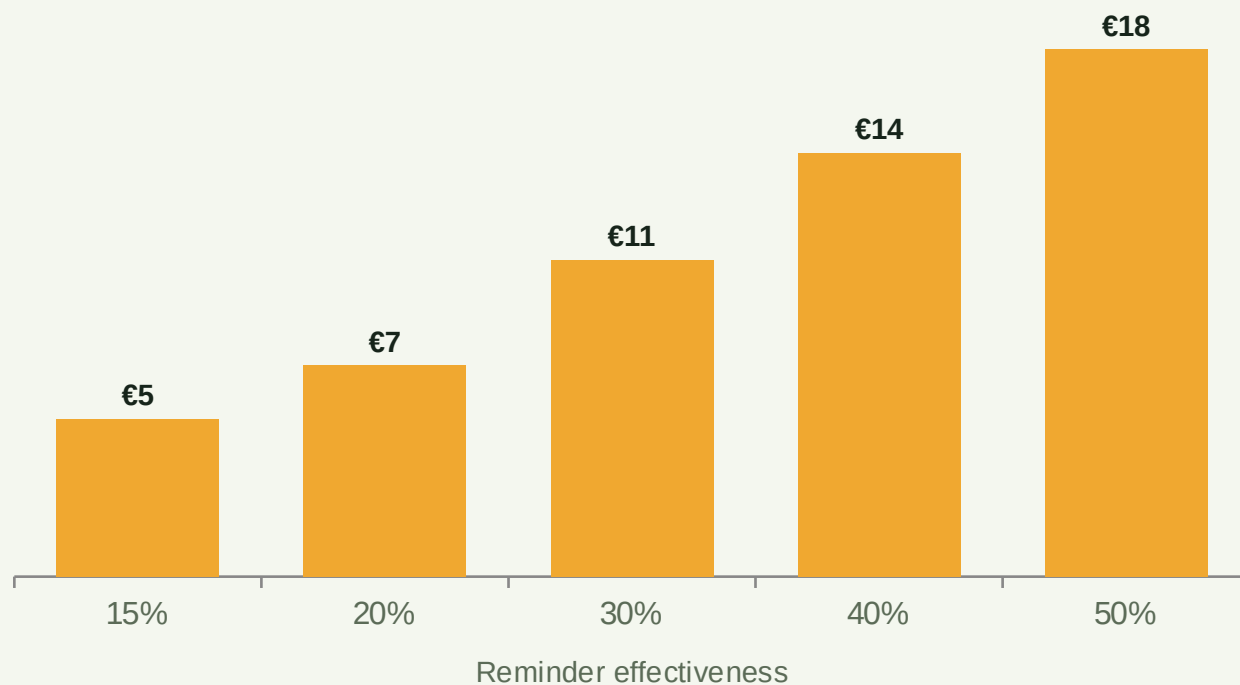
- **Cloudflare Worker** — serves the app AND the API
- **D1 (SQLite)** — accounts, sync, households, community
- **Web Push (VAPID)** — daily expiry reminders
- **Rate-limited auth** — PBKDF2 · 5 tries / 15 min

Deploy: one Cloudflare Worker serves static assets + API on a single origin — installs from static hosting, runs offline.

IMPACT

Projected savings — and how we'll measure it

PROJECTION · MODEL, NOT A STUDY



At a mid-range 30% reminder effectiveness:

≈ **30%** fewer wasted items

≈ **€10-11** saved / household / month

≈ **€128** / household / year

Measured for real by a pilot: a within-subject micro-study (baseline vs. Kulpio week, N = 3-15) using the app's built-in CSV export.

Private by design, hardened by default



Local-first

Your fridge stays in the browser — nothing leaves unless you choose a networked feature.



GDPR export

One tap downloads everything the app holds about you as JSON.



Real deletion

Delete account wipes the server account, sessions & synced data.



Hardened auth

PBKDF2 hashing, opaque tokens, and login rate-limiting (5 / 15 min).

How it evolved — and what I had to solve

01

Offline core

Single-file PWA + shelf-life table.

Challenge solved

Had to run with no network → service-worker precache.

02

Trustworthy dates

AI label reading + estimates.

Challenge solved

Estimates felt unreliable → printed dates always win.

03

Kill the friction

Barcode, receipt scan, paste-a-list.

Challenge solved

Typing was tedious → four fast input paths.

04

Social & safe

Shared fridge, accounts, GDPR, 33 langs.

Challenge solved

Two phones clobbered data → server-side merge.

Where Kulpio goes from here



Roadmap

- Native Android app on the Play Store
- Run the real user pilot & publish measured results
- Validate the offline shelf-life table against reference data
- Email verification (add an email provider)
- Field-level merge for multi-device offline edits
- A shareable monthly impact report



Free & open-source

Released under the GNU AGPL v3 — free for anyone to use, study and build on.

The vision: help households everywhere waste less food and save money — in their own language.



Waste less. Save more. Together.

Kulpio — a food-freshness tracker that fits in your pocket, in your language.

SDG 12

SDG 2

SDG 13

Daniil Bejenari · T.L. Orizont · Ceadîr-Lunga

github.com/Gugster777/kulpio · firSTep Project Competition 2026